Introduction to Computer Architecture and Systems

Chapter 2

Measuring Computers: Why, How, and Takeaways

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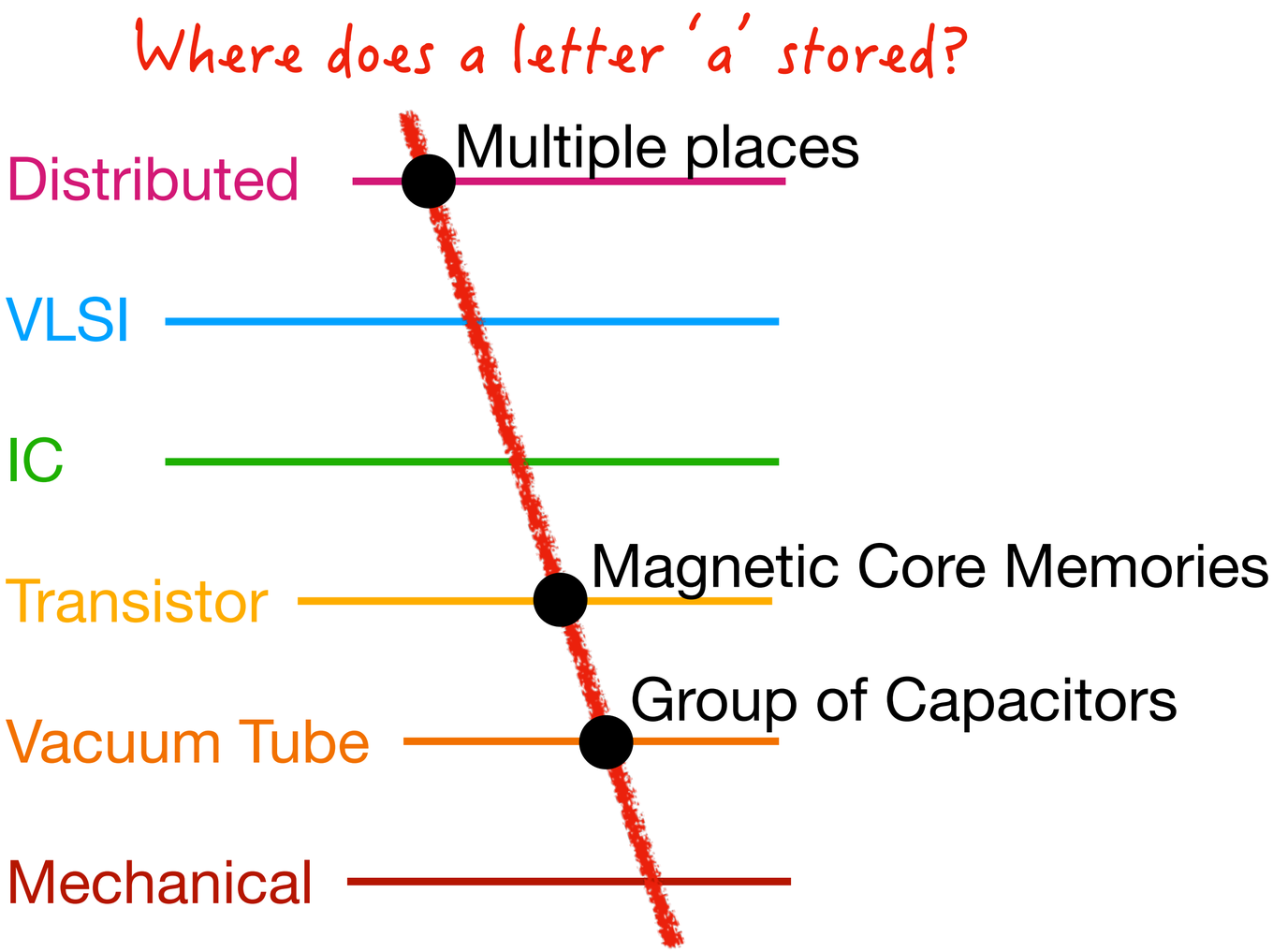
Previously of CSCI 260…

- 6 Stages of Computer's Formation
  - Mechanical - Vacuum Tubes - Transistors - IC - VLSI - 'Invisible'
- What to consider to design a computer?
  - Functional, Non-Functional, Boundaries
- Evaluate CPU's Performance.
  - Intro to CPU Time

Cover-ups…

Building new Metro Lines…

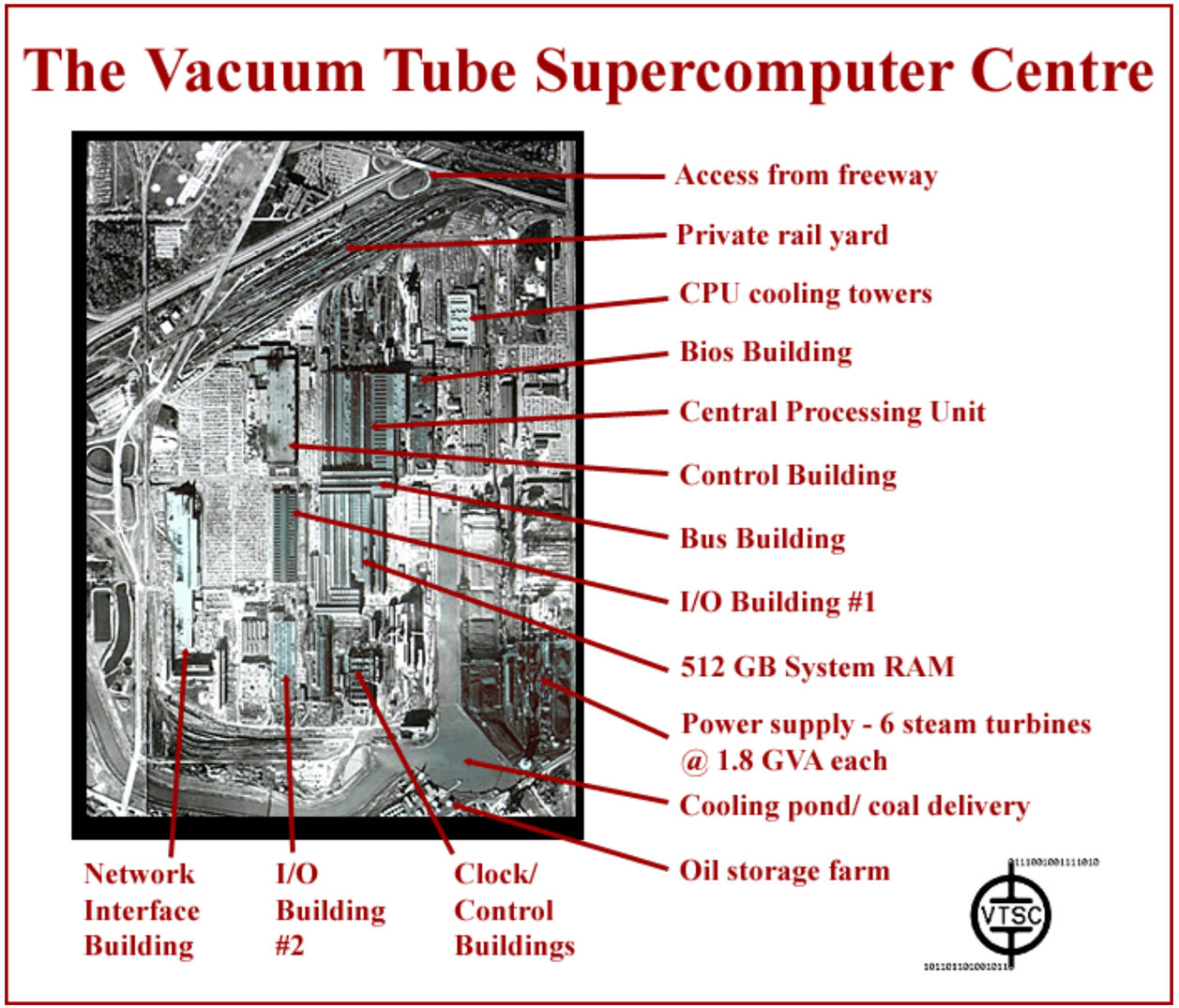
Ask ONE question, and witness it throughout time…





OK,
but why should I get excited?

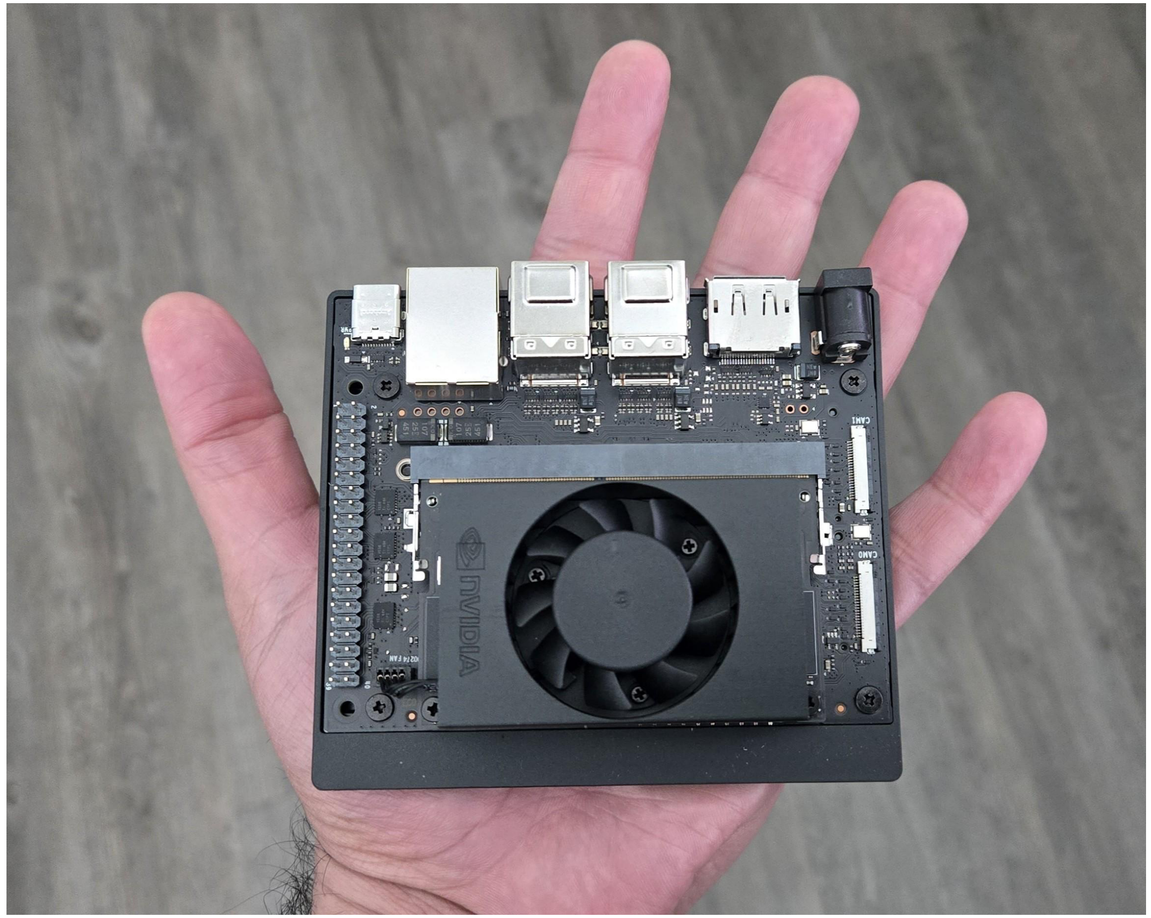
Vacuum Tube Super Computer…



Turning into this…

nVidia Jetson Orin nano w 512 gigs

- Evolution in Computer Architecture gives everyone the power to fast computation, at unparalleled convenient and cost-effective level.
- Most sophisticated structures are created and polished every day, emerging the economy, sciences, and humanities.



Evaluating Computer's Performance (Continue)

CPU Time…

CPU Time

Reverse CPU Time = CPU Performance

$$\text{CPU time} = \text{Instruction Count} \times \text{CPI} \times \text{Clock Cycle Time}$$

$$\text{CPU time} = \frac{\text{Instruction Count} \times \text{CPI}}{\text{Clock Rate}}$$

- Instruction Count (IC)：Total instructions to execute
- CPI (Cycles Per Instruction)：How many clock cycles on average to execute an instruction
- Clock Cycle Time：How much time for a clock cycle（= 1 / Clock Rate）
- Clock Rate：CPU’s frequency（ 3.0 GHz）

CPU Time - Takeaways

Frequency(Clock Cycle Time) does not mean much!

$$\text{CPU time} = \text{Instruction Count} \times \text{CPI} \times \text{Clock Cycle Time}$$

- Instruction Count —Program's Efficiency —Compilers and ISA
- CPI — ISA and Microarchitecture
- Microarchitecture and Technology

Throughput

Throughput Measurement

Rates: Units of work per unit time. Examples:

- millions of instructions / second (MIPS)
- millions of floating point instructions / second (MFLOPS)
- millions of bytes / second (MBytes/sec)
- millions of bits / second (Mbits/sec)
- images / second
- samples / second
- transactions / second (TPS)

MIPS and MFLOPS

MIPS = instruction count / (execution time x 10^6)

MIPS = clock rate / (CPI x 10^6)

MFLOPS - MIPS for floating point operations

- But, MIPS and MFLOPS does not work well every time…
- Think about some scenarios where these criteria are not working.

Counter Example: Video Image Processing

- Number of FLOPS depends on algorithm:

  O(n^2) matrix-vector product vs. O(n log n) FFT

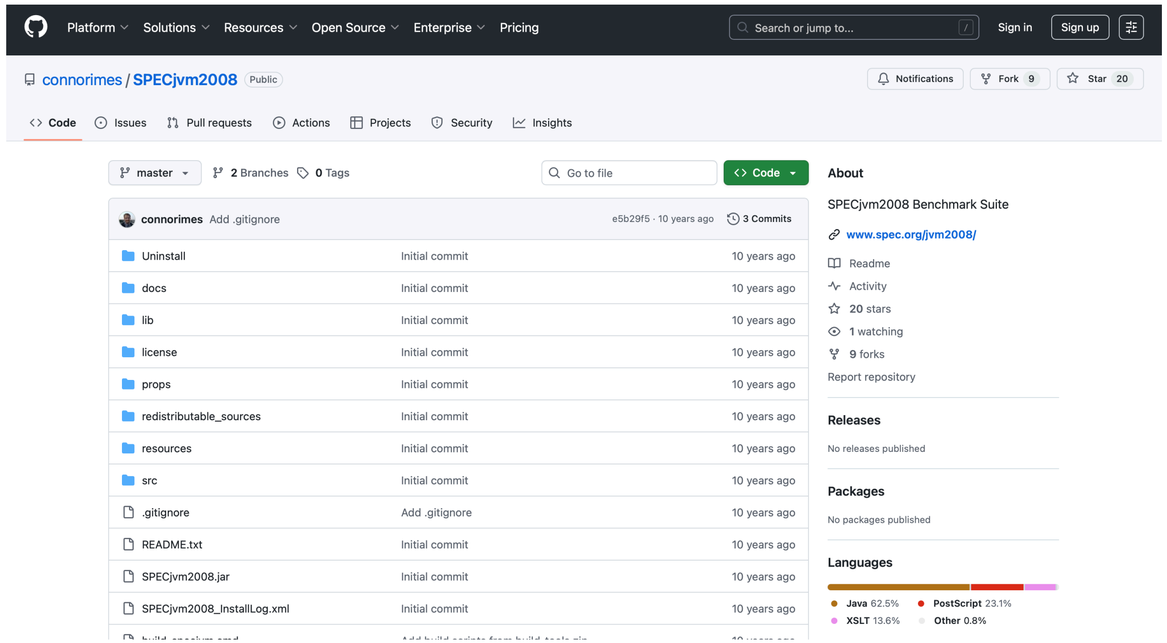
- A Better Criteria: frames/sec:
  - A faster running program will process more frames per second
  - Frames/sec measures speed of target application

Takeaway: Use rates that measure something useful.

Application: Use Benchmark Suites

Ways to standardize the evaluation of machines

- Example: SPEC JVM2008 from Stanford Performance Evaluation Cooperation







Choose the best suite for your domain.

Amdahl’s Law: Parallelism

Amdahl's Law

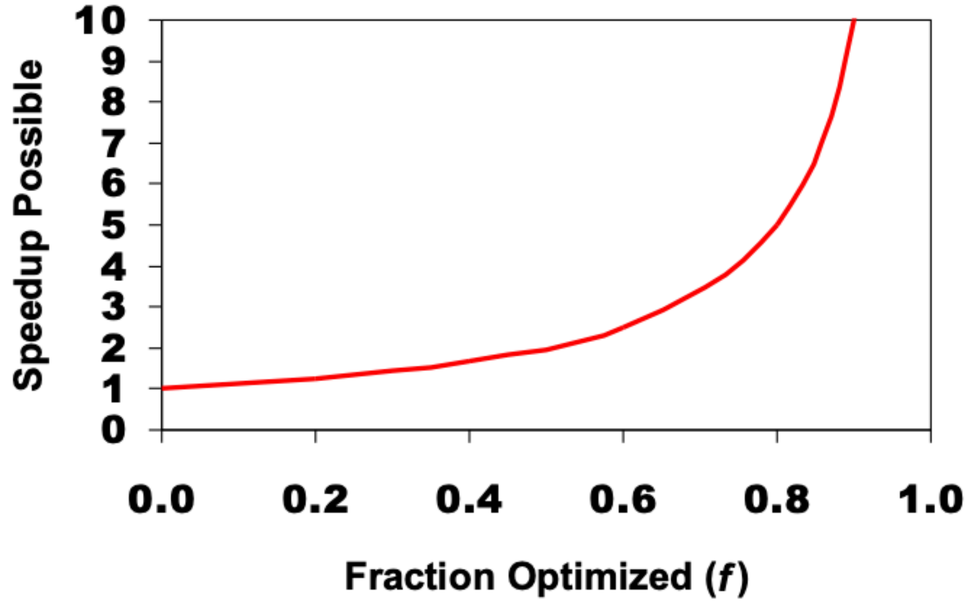
Fraction optimized limits overall speedup

s: Things we can do in parallel

f: Things to do in sequence

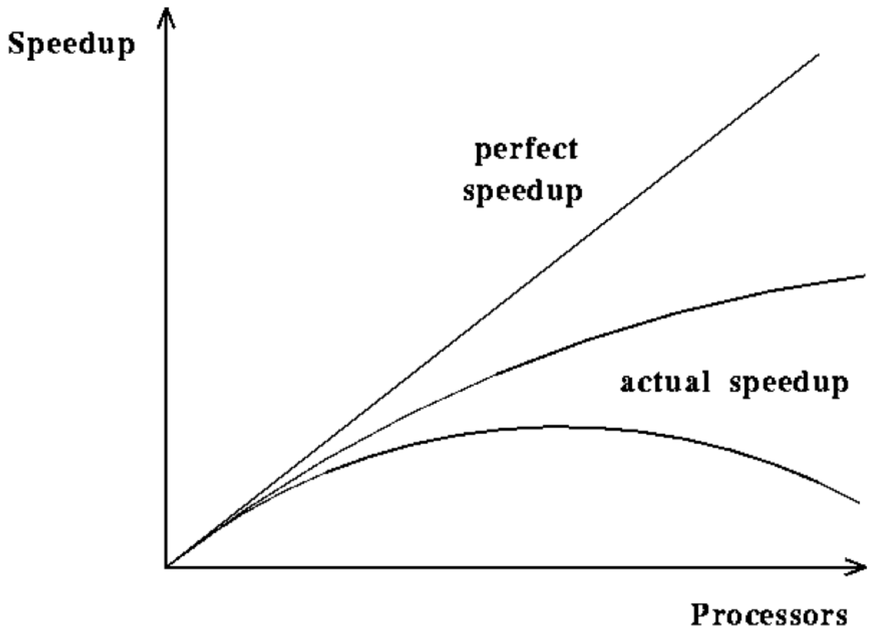
$$Speedup = \frac{1}{1 - f + \frac{f}{s}}$$

where f is fraction optimized,
s is speedup of that fraction



Parallelism

- Parallel Processing - throw more processors at problem
  - 1024 parallel processors - LOTS OF MONEY!
  - 90% of code is parallel (f = 0.9)
  - Parallel portion speeds up by 1024 (s = 1024)
  - Serial portion of code (1-f) limits speedup



Ways of Measuring

Absolute time measure

Straightforward measurement of time a task takes

- running time
- elapsed time
- response time
- latency
- completion time
- execution time

Relative (normalized) time measure

Used to compare machines

- Running time normalized to some reference time
- task_time / reference_time (time = 1 / performance)
- Example:
  - Machine A finishes task in 10 seconds
  - Machine B finishes task in 15 seconds
  - Machine A is (15 seconds / 10 seconds) 1.5x faster than B
  - Machine A is 50% faster than B

Things need to consider…

We talked about MIPS to find out the importance of matching corresponding test suites and scenario. What else?

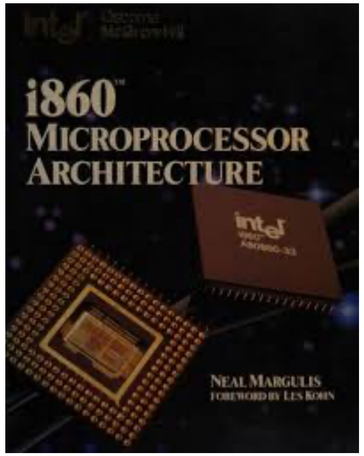
Things to consider: Kernel

Kernel introduces more fluctuations like…

- Context Switch
- Handling Interrupts
- Translation Look-aside Buffer and Cache
- Daemon and Kernel Thread

Things to consider: Peak Rates

- The i860 is advertised as having a peak rate of 80 MFLOPS (40 MHz with 2 flops per cycle). But measuring in the test, we have… sad stories…



Kernel	1D FFT	SASUM	SAXPY	SDOT	SGEMM	SGEMV	SPVMA
MFLOPS	8.2	3.2	6.1	10.3	6.2	15.0	8.1
% of Peak Rates	11%	4%	7%	13%	8%	19%	10%

Don’t trust Peak Rates.

Conclusion

Evaluating CPU

- CPU Time: Contexts and Takeaways
- Amdahl’s Law: Contexts and Takeaways
- Absolute Time Measure and Relative Normalized Measure
- Things need to Consider: Criteria, Kernel, Peak Performance

Next Up:
Labor Day
Bit and Boolean

Why Booleans everywhere…?

Programs are Data

Since the essence of a computer is a Data Manipulation Device

Data is in Binaries...

Since the essence of a computer contains only switches.

... related to Booleans.

Since the essence of a computer is based on conceptual model in Boolean Algebra.

New toy in office: Kenbak-1

Example: KENBAK-1(1978)



Play it online…

Example: KENBAK-1(1978)

Load the program (all octal numbers)

	Location	Contents	Symbolic Instruction
	003	004	Initial value of P Reg.
LOOP	004	103	ADD B C=1
	005	001	
	006	134	STORE B OUTPUT
	007	200	
	010	344	JPD UNC LOOP
	011	004	

OUTPUT	200	xxx	

Steps:

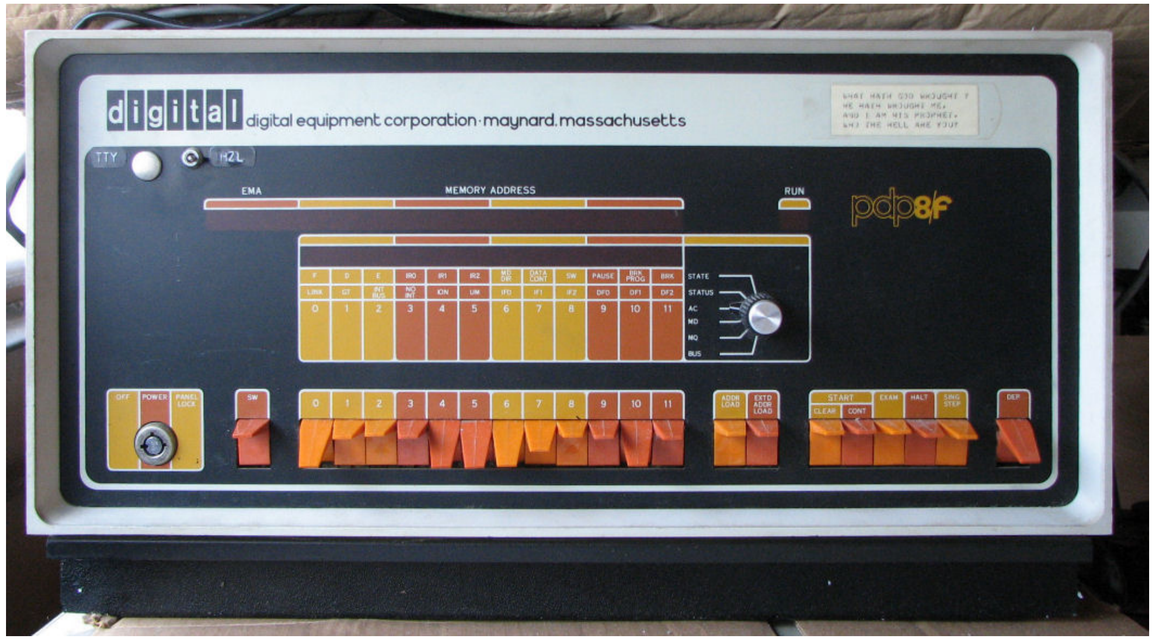
1. Power On
2. Start
3. Stop
4. Clear
5. Enter 003
6. Set (Address)
7. Clear
8. Enter 004
9. Store (Memory)
10. Clear
11. Enter 103
12. Store
13. Clear
14. Enter 001
15. Store
16. Clear
17. Enter 134
18. Store
19. Clear
20. Enter 200
21. Store
22. Clear
23. Enter 344
24. Store
25. Clear
26. Enter 004
27. Store
28. Display (Address)

This program adds 1 to the B Register, puts the value of B out to location 200_8, and hence the Data Lamps, and recycles or loops. The binary counting process should be in evidence on the Data Lamps though the lamps will be operating too fast in the lower order positions to observe the action in detail.

Play it online…

Example: Reference Card

From PDP-8(1965)



- Binary code - Instruction Code
- Code + MUX = Corresponding Logic Cell Fire Ups
- Manipulation of Data = Program Runs

digital

pdp8/i

instruction list

Mnemonic	Code	Operation	Time (μsec.)
		BASIC INSTRUCTIONS	
AND	0000	logical AND	3
TAD	1000	2's complement add	3
ISZ	2000	increment and skip if zero	3
DCA	3000	deposit and clear AC	3
JMS	4000	jump to subroutine	3
JMP	5000	jump	1.5
IOT	6000	in/out transfer	4.25
OPR	7000	operate	1.5
		GROUP 1 OPERATE MICROINSTRUCTIONS (1 CYCLE)	
			Sequence
NOP	7000	no operation	—
CLA	7200	clear AC	1
CLL	7100	clear link	1
CMA	7040	complement AC	2
CML	7020	complement link	2
RAR	7010	rotate AC and link right one	4
RAL	7004	rotate AC and link left one	4
RTR	7012	rotate AC and link right two	4
RTL	7006	rotate AC and link left two	4
IAC	7001	increment AC	3
		GROUP 2 OPERATE MICROINSTRUCTIONS (1 CYCLE)	
			Sequence
SMA	7500	skip on minus AC	1
SZA	7440	skip on zero AC	1
SPA	7510	skip on plus AC	1
SNA	7450	skip on non zero AC	1
SNL	7420	skip on non-zero link	1
SZL	7430	skip on zero link	1
SKP	7410	skip unconditionally	1
OSR	7404	inclusive OR, switch register with AC	3
HLT	7402	halts the program	3
CLA	7600	clear AC	2

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